

experiments.txt

1. HIF-1 α Inhibition in TNBC Cells (~€15,000, 6 weeks) — Cell viability and apoptosis assays after HIF-1 α inhibition | kill if: TNBC cells show no decrease in resistance to chemotherapy
2. Tumor Hypoxia Mimicry (~€18,000, 7 weeks) — Cell viability and metabolic assays under hypoxic conditions | kill if: TNBC cells do not exhibit metabolic reprogramming in response to hypoxia
3. HIF-1 α siRNA Transfection in TNBC Xenografts (~€20,000, 8 weeks) — Tumor growth and chemotherapy response in HIF-1 α knocked-down TNBC xenografts | kill if: Knocked-down HIF-1 α tumors do not show enhanced resistance to chemotherapy
4. Tumor Metabolic Profiling (~€10,000, 5 weeks) — Mass spectrometry-based metabolic profiling of TNBC cells and patient-derived xenografts | kill if: TNBC cells do not exhibit altered metabolism in response to HIF-1 α activation
5. HIF-1 α ChIP-Seq Analysis (~€22,000, 7 weeks) — Chromatin immunoprecipitation sequencing to identify HIF-1 α -bound genes and pathways in TNBC cells | kill if: No significant enrichment of HIF-1 α -bound genes associated with chemotherapy resistance

memo.txt

DECISION SUPPORT MEMO – RESEARCH PRIORITIZATION ONLY

CLAIM UNDER REVIEW:

Tumor cell metabolic reprogramming by HIF-1 α may enhance resistance to chemotherapy in triple-negative breast cancer.

EVIDENCE SUMMARY:

- pubmed_stub | human | 2022 | surrogate | cohort=150

DECISION SUPPORT SCORE: 0.4

ESTIMATED NON-TRANSLATION RISK (PRECLINICAL → HUMAN): 60%

RECOMMENDED ACTION: cheap-test-only

PRIORITIZED CHEAP FALSIFICATION EXPERIMENTS:

1. HIF-1 α Inhibition in TNBC Cells (~€15,000, 6 weeks) — Cell viability and apoptosis assays after HIF-1 α inhibition | kill if: TNBC cells show no decrease in resistance to chemotherapy
2. Tumor Hypoxia Mimicry (~€18,000, 7 weeks) — Cell viability and metabolic assays under hypoxic conditions | kill if: TNBC cells do not exhibit metabolic reprogramming in response to hypoxia
3. HIF-1 α siRNA Transfection in TNBC Xenografts (~€20,000, 8 weeks) — Tumor growth and chemotherapy response in HIF-1 α knocked-down TNBC xenografts | kill if: Knocked-down HIF-1 α tumors do not show enhanced resistance to chemotherapy
4. Tumor Metabolic Profiling (~€10,000, 5 weeks) — Mass spectrometry-based metabolic profiling of TNBC cells and patient-derived xenografts | kill if: TNBC cells do not exhibit altered metabolism in response to HIF-1 α activation
5. HIF-1 α ChIP-Seq Analysis (~€22,000, 7 weeks) — Chromatin immunoprecipitation sequencing to identify HIF-1 α -bound genes and pathways in TNBC cells | kill if: No significant enrichment of HIF-1 α -bound genes associated with chemotherapy resistance

CONFIDENCE DRIVERS:

- + biologically plausible
- + human relevance

CONFIDENCE PENALTIES:

- surrogate endpoint
- limited replication

RELATIVE POSITIONING (PORTFOLIO CONTEXT):

One of several candidate claims generated in this run.

HISTORICAL TRIAGE EXAMPLES (SYSTEM VALIDATION):

Similar claims frequently fail late without early falsification.

DECISION SENSITIVITY:

If ≥ 1 experiment succeeds, escalate. If all fail, terminate this claim.

BOUNDARIES:

Not a treatment recommendation.

Research portfolio decision support only.

experiments.txt

1. Experiment 1: Treg levels in healthy donors vs. melanoma patients (~€12,000, 6 weeks) — Flow cytometry to quantify Tregs in peripheral blood mononuclear cells (PBMCs) | kill if: no significant difference in Treg levels between healthy donors and melanoma patients
2. Experiment 2: In vitro assessment of PD-1 inhibitor efficacy on Tregs (~€18,000, 8 weeks) — Cell culture to evaluate PD-1 inhibitor impact on Treg function and survival | kill if: no significant reduction in Treg suppressive activity after PD-1 inhibitor treatment
3. Experiment 3: Correlation between Treg levels and tumor mutational burden (TMB) (~€25,000, 8 weeks) — Next-generation sequencing to quantify TMB and analyze its correlation with Treg levels | kill if: no significant association between Treg levels and TMB
4. Experiment 4: In vivo assessment of PD-1 inhibitor efficacy in Treg-depleted mice (~€20,000, 6 weeks) — Mouse model to evaluate the impact of Treg depletion on PD-1 inhibitor efficacy | kill if: no significant improvement in PD-1 inhibitor efficacy after Treg depletion
5. Experiment 5: Analysis of public datasets for correlation between Treg levels and treatment outcomes (~€10,000, 4 weeks) — Bioinformatics analysis of publicly available data to identify potential correlations between Treg levels and clinical outcomes | kill if: no significant association between Treg levels and treatment outcomes

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DECISION SUPPORT MEMO – RESEARCH PRIORITIZATION ONLY

CLAIM UNDER REVIEW:

The levels of tumor-infiltrating regulatory T cells (Tregs) may correlate with poor prognosis and reduced response to PD-1 inhibitors in melanoma patients.

EVIDENCE SUMMARY:

- pubmed_stub | human | 2022 | surrogate | cohort=150

DECISION SUPPORT SCORE: 0.5

ESTIMATED NON-TRANSLATION RISK (PRECLINICAL → HUMAN): 60%

RECOMMENDED ACTION: cheap-test-only

PRIORITIZED CHEAP FALSIFICATION EXPERIMENTS:

1. Experiment 1: Treg levels in healthy donors vs. melanoma patients (~€12,000, 6 weeks) — Flow cytometry to quantify Tregs in peripheral blood mononuclear cells (PBMCs) | kill if: no significant difference in Treg levels between healthy donors and melanoma patients
2. Experiment 2: In vitro assessment of PD-1 inhibitor efficacy on Tregs (~€18,000, 8 weeks) — Cell culture to evaluate PD-1 inhibitor impact on Treg function and survival | kill if: no significant reduction in Treg suppressive activity after PD-1 inhibitor treatment
3. Experiment 3: Correlation between Treg levels and tumor mutational burden (TMB) (~€25,000, 8 weeks) — Next-generation sequencing to quantify TMB and analyze its correlation with Treg levels | kill if: no significant association between Treg levels and TMB
4. Experiment 4: In vivo assessment of PD-1 inhibitor efficacy in Treg-depleted mice (~€20,000, 6 weeks) — Mouse model to evaluate the impact of Treg depletion on PD-1 inhibitor efficacy | kill if: no significant improvement in PD-1 inhibitor efficacy after Treg depletion
5. Experiment 5: Analysis of public datasets for correlation between Treg levels and treatment outcomes (~€10,000, 4 weeks) — Bioinformatics analysis of publicly available data to identify potential correlations between Treg levels and clinical outcomes | kill if: no significant association between Treg levels and treatment outcomes

CONFIDENCE DRIVERS:

- + biologically plausible
- + human relevance

CONFIDENCE PENALTIES:

- surrogate endpoint
- limited replication

RELATIVE POSITIONING (PORTFOLIO CONTEXT):

One of several candidate claims generated in this run.

HISTORICAL TRIAGE EXAMPLES (SYSTEM VALIDATION):

Similar claims frequently fail late without early falsification.

DECISION SENSITIVITY:

If ≥ 1 experiment succeeds, escalate. If all fail, terminate this claim.

BOUNDARIES:

Not a treatment recommendation.
Research portfolio decision support only.

experiments.txt

1. miR-21 Inhibitor Effects on NSCLC Spheroids (~€15,000, 6 weeks) — Assay viability and morphology of NSCLC spheroids treated with miR-21 inhibitor, compare to DMSO control. | kill if: No significant change in spheroid morphology or viability.
2. miR-21 Overexpressing NSCLC Cells in Zebrafish Xenografts (~€20,000, 8 weeks) — Transplant miR-21 overexpressing NSCLC cells into zebrafish and monitor invasion and metastasis. | kill if: No significant increase in invasion or metastasis compared to controls.
3. miR-21 Knockdown Effects on NSCLC Cell Migration (~€12,000, 4 weeks) — Use siRNA-mediated knockdown of miR-21 in NSCLC cells and measure migration using transwell assay. | kill if: No significant decrease in cell migration.
4. miR-21 Overexpressing NSCLC Cells in Engineered Lung Tissue (~€18,000, 7 weeks) — Co-culture miR-21 overexpressing NSCLC cells with engineered lung tissue and monitor invasion. | kill if: No significant increase in invasion.
5. miR-21 Inhibitor Effects on NSCLC Tumors in Nude Mice (~€22,000, 8 weeks) — Inject miR-21 inhibitor into established NSCLC tumors in nude mice and monitor tumor growth. | kill if: No

significant decrease in tumor growth.

memo.txt

DECISION SUPPORT MEMO – RESEARCH PRIORITIZATION ONLY

CLAIM UNDER REVIEW:

MicroRNA-21 overexpression may promote epithelial-to-mesenchymal transition and accelerate progression in non-small cell lung cancer.

EVIDENCE SUMMARY:

- pubmed_stub | human | 2022 | surrogate | cohort=150

DECISION SUPPORT SCORE: 0.3

ESTIMATED NON-TRANSLATION RISK (PRECLINICAL → HUMAN): 60%

RECOMMENDED ACTION: cheap-test-only

PRIORITIZED CHEAP FALSIFICATION EXPERIMENTS:

1. miR-21 Inhibitor Effects on NSCLC Spheroids (~€15,000, 6 weeks) — Assay viability and morphology of NSCLC spheroids treated with miR-21 inhibitor, compare to DMSO control. | kill if: No significant change in spheroid morphology or viability.
2. miR-21 Overexpressing NSCLC Cells in Zebrafish Xenografts (~€20,000, 8 weeks) — Transplant miR-21 overexpressing NSCLC cells into zebrafish and monitor invasion and metastasis. | kill if: No significant increase in invasion or metastasis compared to controls.
3. miR-21 Knockdown Effects on NSCLC Cell Migration (~€12,000, 4 weeks) — Use siRNA-mediated knockdown of miR-21 in NSCLC cells and measure migration using transwell assay. | kill if: No significant decrease in cell migration.
4. miR-21 Overexpressing NSCLC Cells in Engineered Lung Tissue (~€18,000, 7 weeks) — Co-culture miR-21 overexpressing NSCLC cells with engineered lung tissue and monitor invasion. | kill if: No significant increase in invasion.
5. miR-21 Inhibitor Effects on NSCLC Tumors in Nude Mice (~€22,000, 8 weeks) — Inject miR-21 inhibitor into established NSCLC tumors in nude mice and monitor tumor growth. | kill if: No significant decrease in tumor growth.

CONFIDENCE DRIVERS:

- + biologically plausible
- + human relevance

CONFIDENCE PENALTIES:

- surrogate endpoint
- limited replication

RELATIVE POSITIONING (PORTFOLIO CONTEXT):

One of several candidate claims generated in this run.

HISTORICAL TRIAGE EXAMPLES (SYSTEM VALIDATION):

Similar claims frequently fail late without early falsification.

DECISION SENSITIVITY:

If ≥ 1 experiment succeeds, escalate. If all fail, terminate this claim.

BOUNDARIES:

Not a treatment recommendation.
Research portfolio decision support only.

experiments.txt

1. miR-155 knockout in ovarian cancer cell lines (~€18,000, 6 weeks) — Western blot, qRT-PCR, and exosome isolation | kill if: no miR-155 expression detected
2. miR-155 mimic in healthy ovarian surface epithelial cells (~€20,000, 7 weeks) — exosome isolation, qRT-PCR, and cell viability assays | kill if: no exosomal miR-155 upregulation detected
3. miR-155 levels in non-cancerous ovarian conditions (~€15,000, 5 weeks) — qRT-PCR and ELISA | kill if: no significant difference in miR-155 levels detected
4. miR-155 expression in response to different chemotherapies (~€22,000, 8 weeks) — qRT-PCR, Western blot, and cell viability assays | kill if: no differential miR-155 expression detected
5. miR-155 levels in ovarian cancer patients with varying response to chemotherapy (~€18,000, 7 weeks) — qRT-PCR and ELISA | kill if: no correlation between miR-155 levels and treatment response detected

memo.txt

DECISION SUPPORT MEMO – RESEARCH PRIORITIZATION ONLY

CLAIM UNDER REVIEW:

Circulating exosomal miR-155 levels may reflect treatment efficacy and predict recurrence of ovarian cancer after chemotherapy.

EVIDENCE SUMMARY:

- pubmed_stub | human | 2022 | surrogate | cohort=150

DECISION SUPPORT SCORE: 0.4

ESTIMATED NON-TRANSLATION RISK (PRECLINICAL → HUMAN): 60%

RECOMMENDED ACTION: cheap-test-only

PRIORITIZED CHEAP FALSIFICATION EXPERIMENTS:

1. miR-155 knockout in ovarian cancer cell lines (~€18,000, 6 weeks) — Western blot, qRT-PCR, and exosome isolation | kill if: no miR-155 expression detected
2. miR-155 mimic in healthy ovarian surface epithelial cells (~€20,000, 7 weeks) — exosome isolation, qRT-PCR, and cell viability assays | kill if: no exosomal miR-155 upregulation detected
3. miR-155 levels in non-cancerous ovarian conditions (~€15,000, 5 weeks) — qRT-PCR and ELISA | kill if: no significant difference in miR-155 levels detected
4. miR-155 expression in response to different chemotherapies (~€22,000, 8 weeks) — qRT-PCR, Western blot, and cell viability assays | kill if: no differential miR-155 expression detected
5. miR-155 levels in ovarian cancer patients with varying response to chemotherapy (~€18,000, 7 weeks) — qRT-PCR and ELISA | kill if: no correlation between miR-155 levels and treatment response detected

CONFIDENCE DRIVERS:

- + biologically plausible
- + human relevance

CONFIDENCE PENALTIES:

- surrogate endpoint
- limited replication

RELATIVE POSITIONING (PORTFOLIO CONTEXT):

One of several candidate claims generated in this run.

HISTORICAL TRIAGE EXAMPLES (SYSTEM VALIDATION):

Similar claims frequently fail late without early falsification.

DECISION SENSITIVITY:

If ≥ 1 experiment succeeds, escalate. If all fail, terminate this claim.

BOUNDARIES:

Not a treatment recommendation.

Research portfolio decision support only.

experiments.txt

1. NK Cell Activation Assay with Osteopontin Inhibition (~€15,000, 6 weeks) — NK cell activation markers (e.g., CD107a) by flow cytometry | kill if: NK cells show no significant decrease in activation markers when treated with osteopontin
2. Osteopontin Knockdown in Glioblastoma Cells and Immune Response Analysis (~€20,000, 8 weeks) — Tumor growth, immune cell infiltration, and cytokine release in a co-culture model | kill if: osteopontin knockdown does not affect tumor growth or immune response
3. Natural Killer Cell Depletion in Glioblastoma Xenografts and Osteopontin Expression Analysis (~€18,000, 7 weeks) — Tumor growth, osteopontin expression, and NK cell infiltration in xenograft models with depleted NK cells | kill if: NK depletion does not affect tumor growth or osteopontin expression
4. Human Osteopontin Variants and NK Cell Activity Comparison (~€12,000, 5 weeks) — NK cell activation by flow cytometry using human osteopontin variants with different binding affinities to NK cells | kill if: osteopontin variants show no difference in NK cell activity
5. Exosome-Mediated Transfer of Osteopontin and Immune Suppression Analysis (~€22,000, 8 weeks) — Immune suppression, osteopontin transfer, and NK cell activation in exosome-treated models | kill if: exosomes containing osteopontin do not suppress immune response

memo.txt

DECISION SUPPORT MEMO – RESEARCH PRIORITIZATION ONLY

CLAIM UNDER REVIEW:

Tumor-derived osteopontin may inhibit natural killer cell activity and contribute to immune evasion in glioblastoma.

EVIDENCE SUMMARY:

- pubmed_stub | human | 2022 | surrogate | cohort=150

DECISION SUPPORT SCORE: 0.5

ESTIMATED NON-TRANSLATION RISK (PRECLINICAL → HUMAN): 60%

RECOMMENDED ACTION: cheap-test-only

PRIORITIZED CHEAP FALSIFICATION EXPERIMENTS:

1. NK Cell Activation Assay with Osteopontin Inhibition (~€15,000, 6 weeks) — NK cell activation markers (e.g., CD107a) by flow cytometry | kill if: NK cells show no significant decrease in activation markers when treated with osteopontin
2. Osteopontin Knockdown in Glioblastoma Cells and Immune Response Analysis (~€20,000, 8

weeks) — Tumor growth, immune cell infiltration, and cytokine release in a co-culture model | kill if: osteopontin knockdown does not affect tumor growth or immune response

3. Natural Killer Cell Depletion in Glioblastoma Xenografts and Osteopontin Expression Analysis (~€18,000, 7 weeks) — Tumor growth, osteopontin expression, and NK cell infiltration in xenograft models with depleted NK cells | kill if: NK depletion does not affect tumor growth or osteopontin expression

4. Human Osteopontin Variants and NK Cell Activity Comparison (~€12,000, 5 weeks) — NK cell activation by flow cytometry using human osteopontin variants with different binding affinities to NK cells | kill if: osteopontin variants show no difference in NK cell activity

5. Exosome-Mediated Transfer of Osteopontin and Immune Suppression Analysis (~€22,000, 8 weeks) — Immune suppression, osteopontin transfer, and NK cell activation in exosome-treated models | kill if: exosomes containing osteopontin do not suppress immune response

CONFIDENCE DRIVERS:

- + biologically plausible
- + human relevance

CONFIDENCE PENALTIES:

- surrogate endpoint
- limited replication

RELATIVE POSITIONING (PORTFOLIO CONTEXT):

One of several candidate claims generated in this run.

HISTORICAL TRIAGE EXAMPLES (SYSTEM VALIDATION):

Similar claims frequently fail late without early falsification.

DECISION SENSITIVITY:

If ≥ 1 experiment succeeds, escalate. If all fail, terminate this claim.

BOUNDARIES:

Not a treatment recommendation.

Research portfolio decision support only.